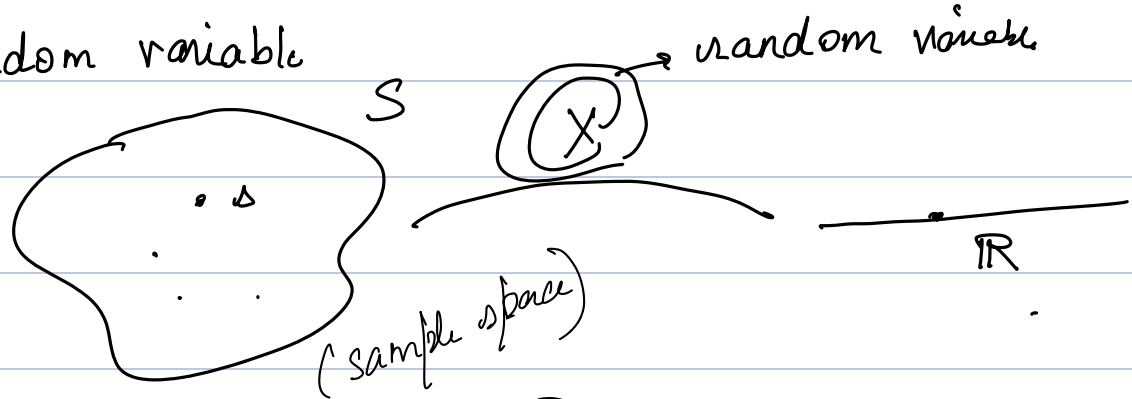
L-4

Random variable



(sample space)

$$X : S \longrightarrow \mathbb{R}$$

$$X(s) \in \mathbb{R}$$

outcome

Ex. $S = \{(HH), (HT), (TH), (TT)\}$

X = # of heads

$$X : S \longrightarrow \{0, 1, 2\}$$

$X(\{HH\}) = 2$;

$X(\{HT\}) = 1$

$X(\{TH\}) = 1$

$X(\{TT\}) = 0$

Y = # of tails

$$Y : S \longrightarrow \{0, 1, 2\}$$

$Y = 2 - X$

$Y(\{HH\}) = 0$

$Y(\{HT\}) = 1$

$Y(\{TH\}) = 1$

$Y(\{TT\}) = 2$

$$Z(s) = \begin{cases} 1 & \text{1st toss is a tail} \\ 0 & \text{otherwise} \end{cases}$$

$Z(\{HH\}) = 0$

$Z(\{TH\}) = 1$

$Z(\{TT\}) = 1$

$Z(\{HT\}) = 0$

Ex. Try to express Z in terms of X & Y

Random Variables

Discrete r.v

Continuous r.v.

Defn: A discrete r.v. X is such that it takes a finite / infinite list of values $a_1, a_2, \ldots$

s.t. $\boxed{\sum P(X = a_j \text{ for some } j) = 1}$

Probability mass function (PMF)

X - discrete r.v

$$p_X(x) = P(X = x)$$

X : # of heads in 2 coin tosses

$$p_X(4) = 0$$

$$p_X(1) = P(\underbrace{X = 1}) = \frac{2}{4} = \frac{1}{2}$$

$\{HT\}, \{TH\}$

$$p_X(2) = P(X = 2) = 1/4$$

$\{HH\}$

$$p_X(3) = 0$$

$$p_X(0) = P(X=0) = 1/4$$

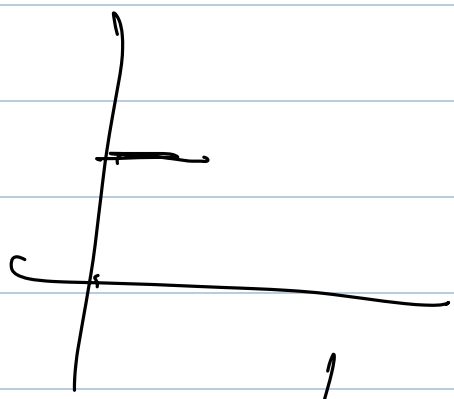
$$\{TT\}$$

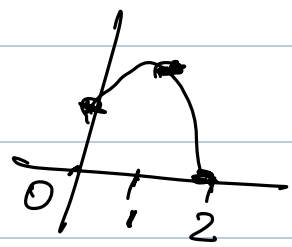
$$\boxed{p_X(x) = \begin{cases} 0 & ; \text{ otherwise} \\ 1/2 & ; x=1 \\ 1/4 & ; x=0,2 \end{cases}}$$

$$p_Y(y) = \begin{cases} 1/4 & y=0,2 \\ 1/2 & y=1 \\ 0 & \text{o/w} \end{cases}$$

Remark: $Y = 2 - X$

$$p_X(x) = p_Y(y)$$





Ex. Roll a fair 6 sided dice twice

$T = X + Y$

$$S = \left\{ \begin{matrix} (1,1), & (1,2) & \cdots & (1,6) \\ (2,1), & (2,2) & \cdots & (2,6) \\ \vdots & & & \end{matrix} \right\}$$

X	Y	$X+Y$
1	1	
2	2	
$\vdots$	$\vdots$	
6	6	

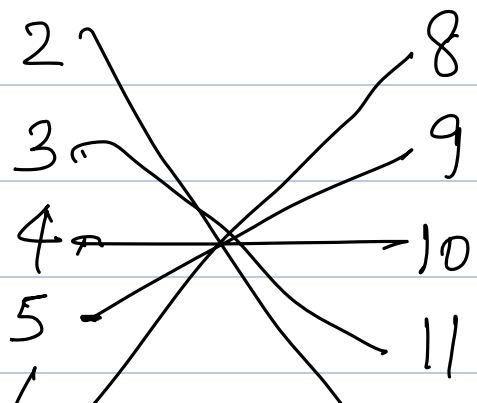
$$P(T=5) = \frac{4}{36} = 1/9$$

$$P(T=11) = 2/36 = 1/18$$

$$P(T=2) = 1/36$$

$$P(T=12) = 1/36$$

$T =$



7 12

$$P(T=7) = 6/36$$

$$P(T=0)=0$$

$$\left\{\begin{array}{l} P(T=2) = P(T=12) = 1/36 \\ P(T=3) = P(T=11) = 2/36 \\ P(T=4) = P(T=10) = 3/36 \\ P(T=5) = P(T=9) = 4/36 \\ P(T=6) = P(T=8) = 5/36 \\ P(T=7) = 6/36 \end{array}\right\}$$

$$P(T=2)+P(T=3)+\cdots+P(T=12) = \underline{1}$$

Thm. $X \to$ discrete r.v ; p_X : PMF of X

(i) $p_X(x) > 0$ if $x = x_j$ for some j and $p_X(x) = 0$ otherwise

$P(X=x) =$ The set of values of $X = \{x_1, x_2, \ldots\}$

(ii) $$\sum_{j=1}^{\infty} p_X(x_j) = 1$$

Previous example:

$T \to \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 \ldots\}$ (x_j)

$p_T(x_j) = P(T = x_j) > 0$ for some x_j

$$\sum_{j=1}^{\infty} p_T(x_j)$$

$$P(T=0) + P(T=1) + P(T=2) + \cdots + P(T=12) + P(T=13) + \cdots = 1$$

Ex $\sum_{j=1}^{\infty} P(X = x_j) = 1 \rightarrow$

L-5

Bernoulli distribution:

A r.v $X \to$ Bernoulli distr with parameter p

$P(X = 1) = p$

$P(X = 0) = 1 - p$

$X \sim Bern(p)$

X

$p_X(x) = P(X = x)$ for any real $x \in \mathbb{R}$

$p_X(x_j) > 0$

$= P(X = x_j) > 0$

for some j range of X

$\{x_1, \ldots, x_{10}\}$

$p_X(x_{11}) = 0$

(i) $p_X(x_j) \geq 0$ for some $1 \leq j \leq 10$

(ii) $\sum p_X(x_j) = 1$

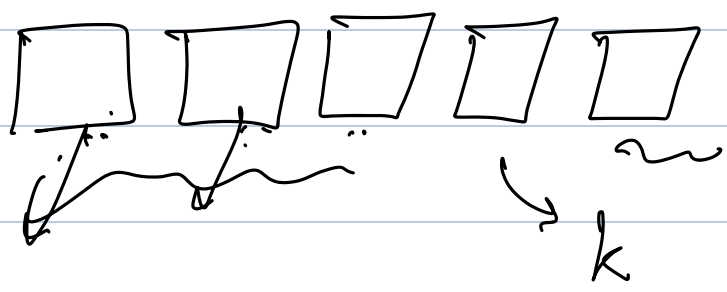
(1) There are 100 prizes, each worth \$1, \$2, ...
You get 5 prizes by picking random boxes, without replacement.
Find PMF of how much your most valuable prize is worth.

X = Value of the most valuable prize among the 5 boxes

Range of X ~~$\{$~~ $\{5, 6, \ldots, 100\}$
for $5 \le k \le 100$, $k \in \mathbb{Z}$

$$p_X(k) = P(X = k) \quad ; 5 \le k \le 100$$

$$= \frac{{}^{k-1}C_4}{{}^{100}C_5}$$



(2) Performing independent Bernoulli trials

N = no. of trials performed until you achieve the 1st success

$p = 1/2$

failure

$$P(N=k) = \underbrace{\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}\cdots\cdots\frac{1}{2}}_{k-1}\cdot\underbrace{\frac{1}{2}}_{k}$$

$$= \left(\frac{1}{2}\right)^k$$

$k =$

M = No. of trials you perform until you get atleast 1 success & atleast 1 failure

PMF of M

$$P(M=m) = \underbrace{\left(\frac{1}{2}\right)^{m-1}\cdot\left(\frac{1}{2}\right)}_{\substack{m-1 \text{ success \&} \\ m\text{th trial} \\ \text{failure}}} + \underbrace{\left(\frac{1}{2}\right)^{m-1}}_{m-1 \text{ failure}}\left(\frac{1}{2}\right)$$

$$= \left(\frac{1}{2}\right)^{m-1}$$

Binomial distribution

Performing n independent Bernoulli trials each with the probability of success $= p$

X = # of success in n trials

$\{0, 1, 2, \ldots, n\}$

$$P(X = k) = \underbrace{{}^nC_k \, p^k (1-p)^{n-k}}$$

$X \sim Bin(n, p)$

Y = # failures in n trials

$= n - X$

(i) $P(Y = k)$

(ii)

$$\sum_{k=0}^{n} P(X = k)$$

$$= \sum_{k=0}^{n} {}^nC_k \, p^k (1-p)^{n-k}$$

$$= (p + 1 - p)^n = 1$$

Remark: "What is the probability distribution of X"

① By listing the probabilities of the values of X in the Range of X

E.g. $P(X=1) =$
$P(X=2) =$
$P(X=3) =$.

② If possible, with a formula that gives the probabability of X

Ex. If X = # of 6's in 10 rolls of a fair die,

$$P(X=k) = \underbrace{{}^{10}C_k \cdot \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{10-k}}_{k=0,1,\ldots,10}$$

③ If possible, with a name of a family of distributions

$$X \sim Bin\left(10, \frac{1}{6}\right)$$

Ex. Toss a coin n times

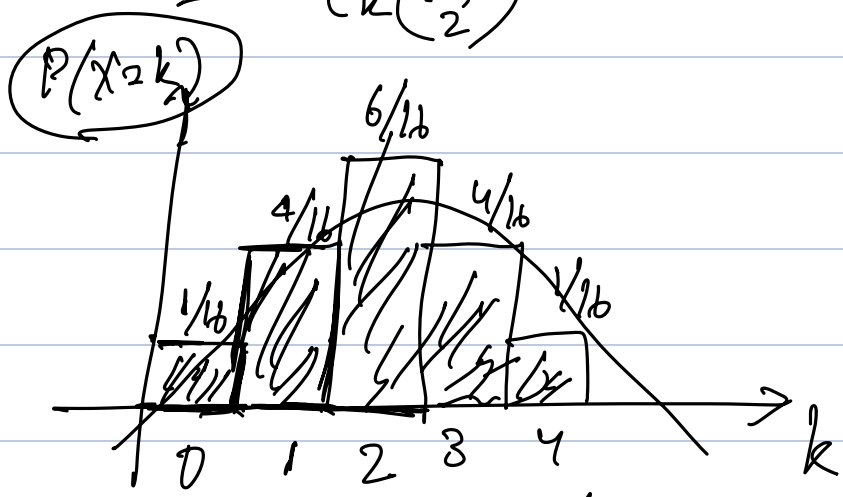
X = # heads (out of n)

$$X \sim Bin\left(n, \frac{1}{2}\right)$$

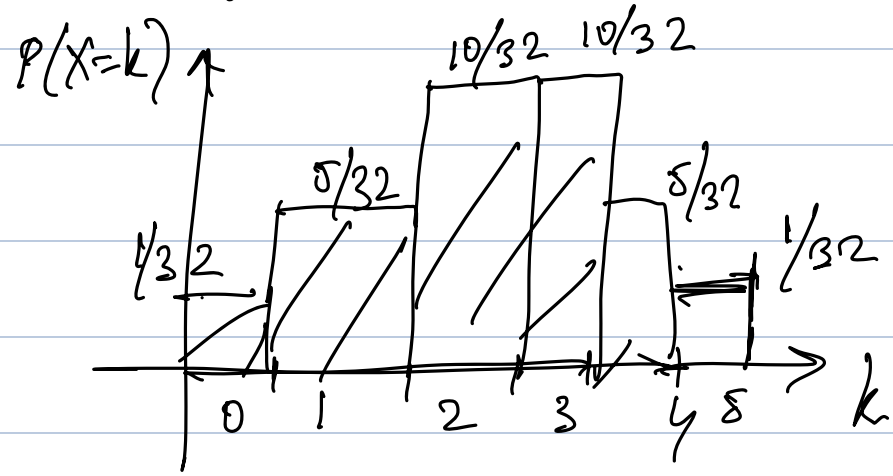
$$P(X=k) = {}^nC_k \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{n-k}$$

$$= {}^nC_k \left(\frac{1}{2}\right)^n$$

$n = 4$



$n = 5$



$$\binom{n}{k} p^k (1-p)^{n-k}$$

Hypergeometric distribution

Consider an urn of w → white balls

b → black balls

Draw n balls at random without replacement

X = no. of white balls drawn out of n many balls

$$X \sim HGeom(w, b, n)$$

$X = k$; $0 \leq k \leq w$

$$P(X=k) = \frac{{}^{w}C_k \cdot {}^{b}C_{n-k}}{{}^{w+b}C_n}$$

$0 \leq k \leq w$, $n \leq b+k$

$n-k \leq b$